

The Journey to a Book Recommendation System

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Abstract

The growing number of books published each year makes it increasingly difficult for readers to discover titles that match their interests. Traditional methods of book discovery, such as bestseller lists, recommendations from friends, or social media trends, often fail to provide personalized suggestions. The objective of this project was therefore to develop and evaluate different recommendation approaches capable of assisting readers in finding books based on their preferences.

Three recommendation systems were implemented and compared. First, a content-based recommender using Term Frequency–Inverse Document Frequency (TF-IDF) was developed based on book metadata from an Amazon books dataset. Second, semantic embeddings generated using a Sentence-BERT model were introduced to capture semantic relationships between books. To improve recommendation quality, a weighted rating score incorporating both average ratings and rating counts was integrated into the ranking process. Finally, a semantic retrieval system was developed using book descriptions from a Goodreads dataset, allowing users to search for books through natural language queries rather than existing book titles.

The results showed that the TF-IDF recommender was able to identify some related books but suffered from limitations associated with keyword matching and dataset inconsistencies. The Sentence-BERT recommender produced more coherent recommendations by considering semantic similarity, while the semantic retrieval system enabled users to search for books based on themes, characters, and story elements. To address the limitation of static datasets, the Google Books API was integrated to retrieve and incorporate newer publications into the recommendation database.

Overall, the findings demonstrate that semantic embeddings and natural language retrieval significantly improve recommendation quality compared to traditional keyword-based approaches. The final system provides a flexible and effective solution for personalized book discovery and highlights the potential of modern Natural Language Processing techniques in recommendation systems.

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Introduction

Reading remains one of the most important cultural and educational activities worldwide. Despite the growing influence of social media, digital platforms, and short-form content, many people continue to spend their leisure time in bookshops and cafés immersed in paperback books. In 2024, 74% of Swiss citizens reported reading as one of their hobbies (Bundesamt für Statistik, 2025). This widespread engagement with reading offers wide-ranging benefits. Beyond providing entertainment, reading can promote relaxation, expand knowledge, and support lifelong learning (Frisch et al., 2025).

But the benefits of reading extend far beyond this. Numerous studies have shown that reading can improve cognitive abilities, increase vocabulary, enhance critical thinking skills, and support lifelong learning (Armstrong et al., 2003; Jenkins et al., 1984; Uchida & Kawashima, 2008). It was also found that many readers associate books with personal development, education, and well-being. In addition, people regularly discuss books with others, demonstrating that reading also contributes to social interaction and community building (Bundesamt für Statistik, 2025).

Considering reading has such a major effect on peoples' lives, it comes without surprise that the publishing industry produces an enormous number of books every year. According to Global Market Statistics, more than 2 million new book titles are released annually, across major publishing markets worldwide (Global Market Statistics, 2026). These publications span a wide variety of genres, including fiction, fantasy, science fiction, romance, mystery, thriller, biography, history, self-help, business, science, and academic literature. The vast number of available books provides readers with unprecedented choice, but it also creates a significant challenge: discovering books that match individual interests and preferences.

For many readers, selecting a book has become increasingly difficult due to the overwhelming amount of available content. Traditional methods of book discovery often rely on browsing bookstores, reading bestseller lists, consulting friends, or searching online reviews. However, these approaches are not always effective. Bookstores often only display books that they assume will make profit, while ignoring others (Pelosi, 2025). Friends can have vastly different tastes or only have a narrow number of books read that you have already all read. Recommendations from friends are often limited because their literary tastes may differ considerably, or their reading experience may be restricted to books that the reader has already read. Lastly, while online reviews can help readers make informed decisions, they are typically consulted after a book has already been selected. As a result, reviews contribute more to the evaluation of books than to the discovery of new titles.

Another common issue is that books are often judged by their covers rather than by their actual content. Publishers invest heavily in cover design because visual appearance strongly influences purchasing decisions (Gudinavičius & Šuminas, 2018). As a result, readers may choose books based primarily on attractive covers, marketing campaigns, or eye-catching titles, even though these factors provide limited information about the

quality of the story itself. At the same time, books with exceptional narratives but less appealing covers may be overlooked.

In recent years, social media platforms have become major drivers of book popularity. Communities such as BookTok (books + TikTok) and Bookstagram (books + Instagram) have significantly influenced book sales and reading trends (Wiederhold, 2022). Here, content creators review and discuss books they have read, share opinions and reading-related content, and recommend books based on titles that readers have previously enjoyed. While these platforms have successfully encouraged many people to read more, they also present certain limitations. Influencers frequently promote books through sponsored partnerships, paid advertising, or publisher collaborations. Consequently, a relatively small number of highly marketed books often receive disproportionate attention, while many equally valuable or potentially more suitable books remain largely undiscovered (da Silva et al., 2025). This creates a recommendation environment that may prioritize visibility and marketing budgets over individual reader preferences and objective quality assessments.

Problem Description and Objectives

The rapid growth of the publishing industry has created an environment in which readers are confronted with an overwhelming number of available books. While this abundance offers greater choice than ever before, it also makes discovering books that align with individual interests increasingly difficult. Traditional methods of book discovery, such as browsing bookstores, relying on bestseller lists, consulting friends, or following recommendations on social media platforms, often fail to provide personalized suggestions that accurately reflect a reader's preferences.

Many existing recommendation sources are influenced by factors unrelated to the actual content of books. Bestseller lists and bookstore displays frequently prioritize commercially successful titles, while social media recommendations are often driven by marketing campaigns, sponsorship agreements, and temporary trends. As a result, readers may be exposed repeatedly to the same highly promoted books while many potentially suitable titles remain undiscovered.

Furthermore, readers typically describe their interests in terms of themes, characters, settings, and story elements rather than specific book titles. For example, a reader may search for "a fantasy novel with dragons, political intrigue, and a strong female protagonist." Traditional recommendation approaches struggle to interpret such requests because they rely heavily on popularity rankings, genres, or manually curated lists.

Therefore, there is a need for an intelligent recommendation system capable of understanding the semantic meaning of both book descriptions and reader preferences. Such a system should identify books based on their content rather than their popularity, marketing exposure, or visual appearance. By leveraging Natural Language Processing

(NLP) and semantic embeddings, readers can receive recommendations that better match their interests and reading intentions.

The main objective of this project was to develop and evaluate a book recommendation system capable of helping readers discover books that match their interests and preferences. To achieve this goal, different recommendation approaches were implemented and compared in order to determine how advances in NLP can improve recommendation quality.

The specific objectives of the project were:

- To develop a baseline content-based recommender using the TF-IDF algorithm and evaluate its ability to recommend books based on metadata such as titles, authors, and genres.
- To improve recommendation quality by incorporating Sentence-BERT embeddings that capture semantic relationships between books rather than relying solely on keyword matching.
- To integrate a weighted rating mechanism that considers both average ratings and rating counts in order to prioritize books that are not only relevant but also well received by readers.
- To investigate the impact of richer textual information by utilizing book descriptions from the Goodreads Best Books dataset.
- To develop a semantic retrieval system that enables users to search for books using natural language queries, such as descriptions of desired themes, characters, or story elements.
- To extend the recommendation database through the integration of the Google Books API and evaluate its suitability for incorporating recently published books.

By achieving these objectives, the project aims to demonstrate how modern NLP techniques can be used to create a more flexible, accurate, and user-friendly book discovery system than traditional recommendation methods.

Theory and Methodology

The development of the book recommendation system followed an iterative approach. Rather than implementing a single solution, three successive recommendation methods were developed and evaluated. Each iteration aimed to address limitations identified in the previous approach and to improve the quality and relevance of the recommendations from the perspective of a typical reader.

This chapter presents the theoretical foundations and methodological considerations underlying the development process. To facilitate a clear understanding of the evolution of the system, the chapter is structured according to the three approaches that were

implemented. For each approach, the motivation behind its selection, the underlying concepts, and the implementation methodology are discussed.

The first approach is based on Term Frequency–Inverse Document Frequency (TF-IDF), a traditional text representation technique commonly used in information retrieval. The second approach utilizes sentence transformer models to capture contextual information within book descriptions. Finally, the third approach employs semantic vector representations generated through a Sentence-BERT (SBERT) embedding model, enabling a more sophisticated understanding of semantic similarities between books. Together, these approaches illustrate the progression from keyword-based matching to advanced semantic recommendation techniques.

Term Frequency-Inverse Document Frequency

TF-IDF is used for representing textual data in information retrieval and natural language processing (Robertson, 2004). The method assigns a numerical weight to each word based on two factors:

1. How frequently the word appears within a document (term frequency)
2. How rare the word is across the entire collection of documents (inverse document frequency)

The primary strength of TF-IDF lies in its simplicity and effectiveness. By converting textual descriptions into numerical vectors, it enables mathematical comparison between documents. Documents containing many of the same important terms will receive similar vector representations and can therefore be identified as related. For this reason, TF-IDF has historically been used in information retrieval systems, search engines, text classification tasks, and recommender systems (Qaiser & Ali, 2018).

In the case of this project, book contents (title, author, main_genre, sub_genre) were used to identify words that were distinct to a certain genre to gain book recommendations for similar-sounding books to the reference book. However, this method represented book titles using a bag-of-words approach, meaning that it only considered the occurrence of words while ignoring their order and grammatical relationships. Further, semantic relationships between different words were ignored. This led to suboptimal results and the need to enhance the book recommendation system through semantic embeddings.

Sentence-BERT Embeddings

Bidirectional Encoder Representations from Transformers (BERT) is a deep learning language model. Unlike traditional text representation methods, BERT considers the context surrounding each word when generating representations (Reimers & Gurevych, 2019). This allows the model to understand that words may have different meanings depending on their context and that different words can express similar concepts.

SBERT extends the original BERT architecture by generating fixed-length embeddings for entire sentences or documents, enabling efficient semantic comparison between texts (Reimers & Gurevych, 2019). In this study, the pre-trained **all-MiniLM-L6-v2** model was

used to generate vector representations of the textual data. The model was selected because it is specifically optimized for semantic search and information retrieval tasks. As a pre-trained model, it could be applied directly to the dataset without requiring additional training or validation. Compared to the original BERT model, which consists of 12 transformer layers, all-MiniLM-L6-v2 contains only 6 layers. This reduced architecture offers a favorable balance between semantic performance and computational cost (Yin & Zhang, 2024).

In a first trial, SBERT was used to recommend books based on the semantic similarities of the book contents. This led to better results than the TF-IDF model. However, the book contents offer very limited information. To further enhance results, book profiles were built (including book descriptions; for more information see chapter data) and converted into a dense numerical vector representing its semantic meaning. Books with similar themes, settings, characters, or story elements are therefore positioned closer together in the embedding space, even if their descriptions contain different vocabulary. Similarity between books was then calculated using cosine similarity, enabling recommendations based on semantic relationships rather than simple keyword matching.

Despite these advantages, the embedding recommender still exhibited an important limitation. It required the reader to provide an existing book title as input. This assumption is not always realistic. Readers often know the type of story they are looking for but cannot name a specific book that matches their interests. This limitation motivated the development of a semantic retrieval system in the next iteration of the project.

Semantic Retrieval System

The third iteration of the recommendation system retained the Sentence-BERT (SBERT) model introduced in the previous approach but fundamentally changed the recommendation architecture. While the earlier embedding-based recommender used SBERT to identify books similar to a selected title, the semantic retrieval system leveraged the same embeddings to process natural-language search queries directly. This transformation shifted the system from an item-to-item recommendation approach to a user-centered semantic search architecture.

The primary architectural change concerned the system input. In the embedding recommender, the user was required to provide the title of an existing book. The system would then retrieve the corresponding embedding and identify books with similar embeddings using cosine similarity. In the semantic retrieval system, this requirement was removed. Instead, readers can enter a free-text description of the type of book they are looking for (Manning, 2008). The system converts this query into an embedding using the same SBERT model and compares it to the embeddings of all books in the dataset.

While this change enhanced the book recommender, it was still limited to the books in the dataset. As this dataset was several years old, many recently published books were not included in the database, limiting the system's ability to recommend current titles. To address this limitation, the Google Books API was integrated into the system as a source of up-to-date book information.

Google Books API

An Application Programming Interface (API) enables software applications to communicate and exchange data automatically. The Google Books API is a web service provided by Google that enables developers to retrieve book-related information directly from Google's book database (Svennerberg, 2010). Through API requests, information such as book titles, authors, descriptions, categories, publication dates, and other metadata can be obtained in a structured JSON format. This information can then be incorporated into existing datasets and used by recommendation algorithms.

Rather than attempting to retrieve thousands of books in a single operation, the system was designed to fetch books incrementally using a small number of targeted queries. For each query, a limited number of results were requested and processed. This approach was chosen because downloading a very large number of books at once would significantly increase processing time. Every newly acquired book must be cleaned, transformed into a book profile, converted into a semantic embedding, and integrated into the existing recommendation database. As the embedding generation process is computationally intensive, especially when using transformer-based language models, processing thousands of new books simultaneously would require substantial time and computing resources.

Rating Scores

One component that remained consistent across the last two iterations of the recommendation system was the use of a hybrid ranking approach. In addition to measuring similarity between books, the system also incorporated reader ratings into the final recommendation score.

To achieve this, a weighted rating score was calculated using both the average rating and the number of ratings a book had received. This approach prevents books with very few ratings from being ranked disproportionately high simply because they have an exceptionally high average score. Instead, books with a large number of ratings are given greater credibility, as their ratings are based on a broader range of reader opinions.

The weighted rating score was normalized and combined with the similarity score to calculate a final ranking score. Consequently, books were ranked not only according to their relevance to the user's query or selected title but also according to their overall reception within the reading community.

Data

For this project, two different datasets were utilized. Both datasets were obtained from Kaggle and served different purposes throughout the development of the recommendation system.

The first dataset, *Books_df* (CHHAVI, 2024) was selected for the development of the initial content-based recommendation system. This dataset contained structured information

such as book titles, authors, genres, sub-genres, average ratings, and rating counts (Table 1). These features provided sufficient information to construct a content-based recommender and allowed the integration of rating-based ranking techniques. However, the dataset contained only limited information about the actual content of the books. As a result, it was not well suited for advanced semantic search and retrieval tasks.

To overcome this limitation, a second dataset (THE DEVASTATOR, 2023), was incorporated into the project. Unlike the Amazon dataset, the Goodreads dataset included book descriptions. These descriptions provided substantially richer textual information, making the dataset suitable for semantic embedding generation and natural-language-based retrieval. Consequently, the Goodreads dataset formed the basis for the semantic embedding recommender and the final semantic retrieval system.

The following sections describe both datasets and the preprocessing procedures applied before model development.

Amazon Books

The Amazon book dataset was uploaded around two years ago by the Kaggle user CHHAVI and it contains 5,740 books (CHHAVI, 2024). Column names and descriptions can be found in Table 1.

Table 1. Columns of Amazon dataset

original column name	new column name	column description
Title	title	The title of the book
Author	author	Name of the author or publication house
Main Genre	main_genre	The main genre the book belongs to
Sub Genre	sub_genre	The specific sub-genre of the book
Rating	average_rating	The average rating of the book given by users
No. Of People rated	ratingsCount	Indicates the count of users who have rated the book
Type	<i>dropped</i>	The format of the book (e.g. paperback or e-book)
Price	<i>dropped</i>	The price of the book in Indian rupees
URLa	<i>dropped</i>	Provides the link to the book's page on Amazon

Before the analysis, several preprocessing steps were carried out. First, columns that were not relevant to the recommendation process were dropped from the dataset (see Table 1). Second, the column names were standardized to improve consistency and simplify later processing (see Table 1). Missing values were not found. Textual attributes were converted to string format to ensure compatibility with NLP techniques. Subsequently, a new column called content was created by combining multiple textual attributes into a single text field. The content feature consisted of the book title, author name, main genre, and sub genre.

Goodreads Best Books

The Goodreads best books dataset was uploaded around three years ago by the Kaggle user THE DEVASTATOR and it contains 49,927 books (THE DEVASTATOR, 2023). Column names and descriptions can be found in Table 2.

Table 2. Columns of Goodreads dataset

original column name	new column name	column description
title	title	The title of the book
series	<i>dropped</i>	The series the book belongs to, if any
author	author	The author of the book
rating	rating	The average rating of the book on GoodReads
description	description	A brief description of the book
language	<i>dropped</i>	The language the book is written in
isbn	<i>dropped</i>	The ISBN number of the book
genres	genres	The genres the book falls into
characters	<i>dropped</i>	The characters featured in the book
bookFormat	<i>dropped</i>	The format of the book (e.g paperback or e-book)
edition	<i>dropped</i>	The edition of the book
pages	<i>dropped</i>	The number of pages in the book
publisher	<i>dropped</i>	The publisher of the book
publishDate	<i>dropped</i>	The date the book was published
firstPublishDate	<i>dropped</i>	The date the book was first published
awards	<i>dropped</i>	Any awards the book has received
numRatings	numratings	The number of ratings the book has received
ratingsByStars	ratingsbystars	The ratings the book has received broken down by star ratings
likedPercent	likedpercent	The percentage of readers who liked the book
setting	<i>dropped</i>	The setting of the book
coverImg	<i>dropped</i>	The cover image of the book
bbeScore	<i>dropped</i>	The score of the book on GoodReads' Best Books Ever list
bbeVotes	<i>dropped</i>	The number of votes for the book on GoodReads' Best Books Ever list
price	<i>dropped</i>	The price of the book

For this dataset, the same preprocessing steps were performed. The columns that were not relevant to the recommendation process were dropped from the dataset (see Table 2). Then the column names were standardized as well (Table 2). Missing values were not found via the `books.isnull()` function. However, looking into the dataset, several book descriptions were misaligned. Despite the dataset appearing inconsistent, a definitive solution to resolve these issues could not be implemented. Since the remaining components of the system functioned as intended, the dataset was ultimately used in its existing form, accepting the limitation that certain books may not be recommended accurately due to incomplete or irregular data entries.

To create a comprehensive textual representation of each book, a new variable called *bookprofile* was generated. This variable combined multiple descriptive attributes into a single text field. Depending on availability, the book profile included the title, genre information, and the book description.

The book profile served as the primary input for the semantic embedding model. By combining several sources of information into a single text representation, the model was able to capture not only genre information but also story elements, themes, character types, and settings described within the book summaries.

Results

TF-IDF Content-Based Recommender

The first recommender system developed in this project was a content-based recommender using the TF-IDF approach. To evaluate the system, the book *Fantastic Beasts and Where to Find Them* was selected as an example. This title was chosen because it is both a well-known book series and a spin-off of the highly successful Harry Potter franchise. Therefore, a functioning recommendation system should ideally return books from the same series or books closely related to the Harry Potter universe.

The recommendation results showed a mixed performance (see Figure 1). The first two recommendations and the fifth recommendation were part of the Fantastic Beast franchise and therefore represented successful recommendations. However, the third and fourth recommendations differed substantially from the expected output, as they were unrelated to any fantasy books.

Figure 1. Results of TDF-IDF recommender

Recommendations:

	title	author \
6	Fantastic Beasts: The Crimes of Grindelwald - ...	J.K. Rowling
23	FANTASTIC BEASTS: THE SECRETS OF DUMBLEDORE İç...	J.K. Rowling
9	Screenplay	Syd Field
40	Will	Will Smith
33	The Magic of MinaLima: Celebrating the Graphic...	MinaLima

	main_genre
6	Arts, Film & Photography
23	Arts, Film & Photography
9	Arts, Film & Photography
40	Arts, Film & Photography
33	Arts, Film & Photography

This behaviour can be explained by the underlying bag-of-words principle used by TF-IDF. The algorithm evaluates the occurrence of words across all books and assigns weights based on their frequency (Manning, 2008). Importantly, the algorithm does not only consider which words occur in two books, but also which words do not occur. As a result, books with very short titles or limited textual information occasionally received unexpectedly high similarity scores (Schütze et al., 2008).

A further issue became apparent when examining the genres assigned to the recommended books. All recommendations were classified under the main genre “Arts, Film & Photography”. While all recommendations could arguably be associated to film, grouping a fantasy children’s book, a screenplay writing guide and an autobiography under the same genre appears questionable from a recommendation perspective. However, this is less a reflection on the performance of the recommender but rather a limitation of the dataset.

Another limitation of the TF-IDF approach is that semantic meaning is ignored. The model only considers exact word occurrences and cannot understand that different words may describe similar concepts. Therefore, books were recommended primarily because they shared words with the query title rather than because they contained similar stories, themes, or characters.

SBERT Embeddings Recommender with Weighed Ratings

Unlike TF-IDF, Sentence-BERT converts textual information into dense vector representations that encode semantic relationships between words and sentences. Consequently, books no longer need to share identical words to be considered similar (Reimers & Gurevych, 2019). In addition, a weighted rating component was incorporated into the recommendation process to reduce the influence of books with artificially high ratings based on only a small number of reviews.

For this experiment, ten recommendations instead of five were generated for the same Fantastic Beasts book (Figure 2). It was expected that due to the increased number of recommendations, not only Fantastic Beasts books would be recommended but also from the broader Harry Potter universe.

Figure 2. Results of SBERT recommender

Recommendations:			
	title	author	\
6	Fantastic Beasts: The Crimes of Grindelwald - ...	J.K. Rowling	
23	FANTASTIC BEASTS: THE SECRETS OF DUMBLEDORE İç...	J.K. Rowling	
33	The Magic of Minalima: Celebrating the Graphic...	Minalima	
100	Harry Potter and the Cursed Child - Parts One ...	J.K. Rowling	
900	Harry Potter and the Chamber of Secrets	J.K. Rowling	
962	Harry Potter and the Deathly Hallows	J.K. Rowling	
897	Harry Potter and the Philosopher's Stone	J.K. Rowling	
961	Harry Potter and the Order of the Phoenix	J.K. Rowling	
7737	Harry Potter - The Illustrated Collection	J. K. Rowling	
903	Harry Potter and the Goblet of Fire	J.K. Rowling	
	main_genre	average_rating	ratingsCount
6	Arts, Film & Photography	4.6	10152
23	Arts, Film & Photography	4.6	1280
33	Arts, Film & Photography	4.8	595
100	Arts, Film & Photography	4.4	86578
900	Children's Books	4.7	86490
962	Children's Books	4.8	87027
897	Children's Books	4.7	107802
961	Children's Books	4.7	66771
7737	Teen & Young Adult	4.8	691
903	Children's Books	4.8	67626

The results confirmed this expectation. All ten recommended books were directly related to the Harry Potter franchise or its associated universe. Compared to the TF-IDF recommender, the recommendations were substantially more coherent and relevant.

Interestingly, the main genre differed for these books, even though they are all from the same connected universe.

Furthermore, the effectiveness of the weighted rating approach could be observed in the ranking order. Books with slightly lower average ratings (approximately 4.6 stars) occasionally appeared above books with higher average ratings (approximately 4.8 stars) because they had significantly larger numbers of ratings. This indicates that the ranking algorithm successfully balanced quality and popularity rather than relying solely on average ratings.

Overall, the SBERT recommender produced more meaningful recommendations than the TF-IDF system and demonstrated the benefits of incorporating semantic similarity and reader feedback into the recommendation process.

Semantic Retrieval System & New Dataset

Although the SBERT recommender produced substantially improved results, it was still limited by the information available in the original Amazon dataset. Since the dataset primarily contained titles, authors, and genre information, the model had limited knowledge about the actual content of the books.

To address this issue, the Goodreads Best Books dataset was introduced. This dataset contains detailed book descriptions, allowing the embedding model to learn information about plots, themes, characters, settings, and story elements. After generating semantic embeddings from the Goodreads descriptions, recommendations were produced again using the same example book. As expected, the results differed slightly from those obtained using the Amazon dataset.

The first recommendation was essentially the same book, only listed under a shorter title variation and therefore not excluded by the recommendation process. The second recommendation corresponded to the same recommendation observed in the previous experiment, despite the use of a completely different dataset. This finding is particularly interesting because it suggests that the recommendation was driven by genuine content similarity rather than dataset-specific characteristics.

Overall, the recommended books remained closely related to the Fantastic Beasts and Harry Potter universe. However, the genre classifications appeared more appropriate and elaborate than those observed in the Amazon dataset, indicating that the Goodreads data provided a more accurate representation of book content.

Because the researcher was already familiar with the recommended books, it was relatively easy to verify that the recommendations were relevant. However, a practical recommendation system should not only suggest books that the user already knows. Instead, it should allow users to describe desired characteristics and receive recommendations for previously unknown books. This requirement motivated the development of the final semantic retrieval system.

Semantic Retrieval Through Natural Language Queries

The final stage of the project involved the development of a semantic retrieval system capable of processing natural language queries. Instead of entering a book title, users can describe the type of book they wish to read. For example, the query "A fantasy book with dragons and a female main character" was used to evaluate the system.

The retrieved books matched the requested characteristics remarkably well. Inspection of the results confirmed that the recommended books featured fantasy settings, dragons, and female protagonists (Figure 3). Unlike the previous recommendation approaches, the retrieval system was no longer limited to finding books similar to a specific title. Instead, it could identify books based on thematic and narrative characteristics described directly by the user.

This significantly increased the flexibility and practical usefulness of the system. Readers are no longer required to know an existing book before receiving recommendations. Instead, they can search directly for themes, characters, or story elements that interest them.

Figure 3. Results of first semantic retrieval

	title \			
31268	Dragonsong / Dragonsinger			
6216	A Natural History of Dragons			
22757	Loved by a Dragon			
10592	Dragonflight / Dragonquest			
2549	The Dragonriders of Pern			
1978	Dragons of Autumn Twilight			
25146	Dragon Lyric: A Short Story			
13355	Dragon Keeper			
47302	Playing With Fire			
11905	Searching for Dragons			
		author	rating	numratings \
31268		Anne McCaffrey	4.56	1930
6216		Marie Brennan (Goodreads Author)	3.80	25894
22757		Linda K. Hopkins (Goodreads Author)	4.25	459
10592		Anne McCaffrey, Adrienne Barbeau (Read by), Mi...	4.47	10906
2549		Anne McCaffrey	4.24	27108
1978		Margaret Weis (Goodreads Author), Tracy Hickma...	3.99	108100
25146		Bethany A. Jennings (Goodreads Author)	4.37	70
13355		Carole Wilkinson	3.99	10633
47302		Katie MacAlister (Goodreads Author)	4.00	8640
11905		Patricia C. Wrede (Goodreads Author), Peter de...	4.26	41008

However, one limitation became apparent. Many retrieved books were relatively old. Since the Goodreads dataset was collected several years ago, recently published books were not represented in the recommendation database. Consequently, newer titles could not be retrieved regardless of their relevance to the user's query. In addition, the dataset itself is described online as entailing the best rated goodreads books of all time, meaning that by default, this dataset probably had more older than younger books, as they could have been rated over years while newer books had less time on the platform. Again, this does not show a limitation of the retrieval system but a possible limitation of the dataset itself.

Google Books API Integration

To address the issue of outdated recommendations, the Google Books API was integrated into the system.

The API was used to retrieve more recent books and enrich the recommendation database. Initial experiments revealed several challenges. Although each API request successfully returned up to 40 books, many retrieved records contained incomplete metadata, particularly missing descriptions. Since descriptions are essential for generating semantic embeddings, only a small proportion of the retrieved books could be incorporated into the system.

Another practical consideration involved API usage limits. Attempting to retrieve thousands of books in a single operation would not only increase computational requirements but could also exceed the API quota and potentially incur additional costs.

For this reason, a targeted retrieval strategy was adopted. Instead of requesting large numbers of random books, the API was queried using specific genres such as “romantasy”, “high fantasy”, “dragons”, “wyvern”, “war college”, and “new adult” (results see Figure 4). Furthermore, only books published from approximately 2023 onwards were targeted, as older books were already expected to be present within the Goodreads dataset.

Figure 4. Books fetched via Google API

Books fetched: 600			
	title	author	\
0	Romantasy	Isabella Mey	
1	Dead Souls: Die spicy Romantasy-Trilogie in ei...	Izzy Maxen	
2	Infernas 1: King of Ash	Melanie Lane	
3	I Am the Blade	Marie Graßhoff	
4	The dark between us	Lily Cattens	
	description	genres	publishedDate
0	Romantasy ... ist ein Inselkönigreich auf dem ...	Fiction	2024-02-28
1	»Ich darf ihn nicht wiedersehen wollen. Ich da...	Fiction	2024-11-07
2	»Und?«, raunte Dante leise, den Blick auf mein...	Fiction	2023-11-24
3	Dieses Buch gibt es in zwei Versionen: mit und...	Fiction	2026-06-29
4			2026
Books fetched: 193			

Using this strategy, approximately 600 additional books were retrieved and processed, with 193 books having descriptions. Inspection of the retrieved records confirmed that the books belonged to the intended genres. This demonstrates that the Google Books API can successfully supplement the original dataset.

However, when the semantic retrieval system was tested again, the recommendations remained largely unchanged compared to the results obtained without the additional Google Books data. One possible explanation is that the example query was not sufficiently related to the newly retrieved books, meaning that the added titles did not rank highly enough to appear among the top recommendations.

To investigate this further, a second query was used: *“a romantasy new adult book with dragons and a female main character.”* Despite being more specific and closely aligned with some of the genres used during data collection, the recommendations remained similar to those generated by the original query. This suggests that the existing Goodreads dataset already contained a large number of highly relevant books for these search terms, reducing the influence of the newly added titles.

Although the inclusion of Google Books data did not significantly improve the recommendations in the conducted experiments, the approach still offers considerable potential. By retrieving larger numbers of books or targeting more specific genres, themes, or publication periods, the API could provide valuable additions to the recommendation database. This would be particularly beneficial for niche interests or for ensuring that recently published books are represented within the retrieval system.

Conclusion

The aim of this project was to develop and evaluate different approaches for recommending books based on their content. To achieve this, three recommendation systems of increasing complexity were implemented: a TF-IDF content-based recommender, an SBERT embedding recommender with weighted ratings, and a semantic retrieval system capable of processing natural language queries.

The TF-IDF recommender provided a useful starting point and was able to identify some related books. However, the results also revealed the limitations of keyword-based approaches, as recommendations were influenced by exact word matches and inconsistencies within the dataset. To overcome these limitations, SBERT embeddings were introduced. By representing books as semantic vectors, the recommender was able to identify books with similar meanings rather than simply matching words. The integration of weighted ratings further improved the quality of recommendations by favouring books that were both relevant and well received by readers.

To enhance semantic understanding, the Goodreads dataset was used because it contained detailed book descriptions. This additional information enabled the development of a semantic retrieval system that allows users to search for books using natural language descriptions instead of existing book titles. The results demonstrated that the system was able to retrieve books matching specific themes, characters, and story elements. Finally, the Google Books API was integrated to supplement the dataset with newer publications. This addressed the limitation of relying solely on older datasets and improved the system's ability to recommend recently released books.

Overall, the project demonstrated that recommendation quality improved significantly when moving from traditional keyword-based methods to semantic embeddings and natural language retrieval. The final system provides a flexible and effective approach to helping readers discover books that match their interests and preferences.

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